

Question:

Shock causes:

Answer Choices:

- A. Stagnant hypoxia
- B. Anemic hypoxia
- C. Hypoxic hypoxia
- D. Histotoxic hypoxia

Reasoning + Answer:

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Shock leads to inadequate tissue perfusion due to impaired blood flow , resulting in insufficient oxygen delivery to tissues . This is primarily due to reduced oxygen content in the blood because of poor circulation , not because of a defect in oxygen binding or utilization .

In hypoxic hypoxia , there is a reduced oxygen partial pressure in the alveoli or blood , leading to low oxygen content in the blood . In shock , especially hypovolemic or cardiogenic shock , the blood flow to tissues is diminished , which directly reduces the delivery of oxygen to cells . This condition is classified as hypoxic hypoxia because the tissue is not receiving enough oxygen due to poor perfusion .

Stagnant hypoxia occurs when blood flow is reduced but oxygen is still present , and the tissue is not receiving adequate oxygen due to poor circulation , which is exactly what happens in shock . However , the term "stagnant hypoxia " is more specifically used when there is a lack of blood flow due to vascular obstruction or reduced cardiac output , and the tissue is not receiving oxygen despite adequate oxygen content in the blood .

Anemic hypoxia occurs when there is a decrease in hemoglobin or red blood cells , leading to reduced oxygen -carrying capacity . This is not the primary mechanism in shock , though it may occur secondarily .

Histotoxic hypoxia involves impaired oxygen utilization by cells (e.g ., due to cyanide poisoning), which is not relevant in typical shock .

Therefore , the most accurate answer is hypoxic hypoxia because in shock , the primary issue is reduced oxygen delivery due to poor perfusion , which directly corresponds to hypoxic hypoxia .

< answer >

C. Hypoxic hypoxia

</answer>